# Integrated Network Pharmacology and Molecular Dynamics Reveal Multi-Target Mechanisms of Cucurbitacin C7 Against Prostate Cancer

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**Draft notice.** Red author queries identify information that must be verified from laboratory records or source files before submission. They are not intended to appear in the submitted manuscript.

**Abstract**

Prostate cancer remains a major cause of cancer-related morbidity and mortality, and available treatments are often limited by resistance, toxicity, and disease recurrence. Cucurbitacin C7 (CuC7), a cucurbitacin C derivative, has shown cytotoxic activity, but its potential molecular associations in prostate cancer remain insufficiently characterized. We combined network pharmacology, molecular docking, 400-ns molecular dynamics (MD) simulations, molecular mechanics/Poisson–Boltzmann surface area (MM/PBSA) calculations, and in vitro assays in DU145 cells. Thirty-two proteins overlapped between predicted CuC7 targets and prostate cancer-associated genes. Network analysis prioritized AKT1, EGFR, MDM2, PTGS2, PIK3CA, BCL2L1, and ERBB2, and enrichment analysis implicated apoptosis, phosphorylation, and PI3K–AKT, MAPK, FoxO, and EGFR-related signaling. Among these seven proteins, PTGS2 produced the most negative CuC7 docking score (−8.73 kcal/mol). Redocking of co-crystallized ligands across all seven targets yielded heavy-atom root mean square deviations (RMSDs) of 0.186–1.490 Å. Comparative MD and MM/PBSA analyses identified AKT1 as a prominent candidate, with an estimated CuC7 binding energy of $-59.36 \pm 3.00$ kcal/mol. CuC7 reduced DU145 cell viability in a concentration-dependent manner ($IC_{50} = 1.787$ $\mu$M; 95% CI, 1.654–1.928 $\mu$M) and was associated with reduced AKT1 protein levels. These findings support a testable multi-target hypothesis for CuC7 in prostate cancer; however, direct binding, target engagement, and AKT1 kinase inhibition remain to be demonstrated experimentally.

**Keywords:** Cucurbitacin C7; prostate cancer; network pharmacology; molecular docking; molecular dynamics; AKT1; PI3K–AKT pathway

# 1 Introduction

Prostate cancer (PCa) is one of the most prevalent malignancies affecting men worldwide and remains a major contributor to cancer-related morbidity and mortality (Siegel et al., 2021; Sung et al., 2021). Despite significant advances in surgical, hormonal, and chemotherapeutic approaches, current treatment strategies are often limited by drug resistance, systemic toxicity, and disease recurrence (Teo and Rafiq, 2021; Wang et al., 2018). These challenges underscore the urgent need to identify novel, effective, and safer therapeutic agents.

Natural products derived from plants and agricultural residues have emerged as a valuable source of bioactive compounds with anticancer potential. Cucumber (*Cucumis sativus* L.) generates substantial agricultural by-products, including leaves, stems, peels, seeds, and juice residues, which are often discarded despite being rich in phytochemicals. Among these metabolites, cucurbitacin C (CuC), a tetracyclic triterpenoid, along with phenolic acids and flavonoids, has demonstrated antioxidant, anti-inflammatory, and anticancer activities. Notably, cucurbitacin C is differentially distributed within the plant, with higher concentrations reported in leaves and stem tips compared with fruits, roots, and seeds (Bauleth et al., 2025a; Horie et al., 2007; Wu et al., 2019a). This highlights the potential value of agricultural waste as a sustainable source of pharmacologically active compounds.

Previous studies have shown that CuC exerts anticancer effects in prostate cancer by inhibiting cell proliferation, inducing cell-cycle arrest, activating caspase-dependent apoptosis, suppressing the Akt signaling pathway, and downregulating cyclins. However, its cytotoxicity varies across prostate cancer cell lines, with DU145 cells showing moderate sensitivity ($IC_{50} \approx 8\ \mu M$), while PC-3 ($IC_{50} \approx 0.02\ \mu M$) and LNCaP ($IC_{50} \approx 0.16\ \mu M$) cells are more sensitive (Qing et al., 2022a; Wu et al., 2019b). These differences highlight the need for more potent and structurally optimized derivatives to improve therapeutic efficacy.

Qing et al. (2022a) reported the isolation of ten cucurbitacin C-type analogues from cucumber leaves, including seven previously undescribed derivatives. Among the ten analogues evaluated against DU145 cells, cucurbitacin C7 (CuC7) exhibited the greatest cytotoxic activity ($IC_{50} = 1.82 \pm 0.27\ \mu M$), corresponding to an approximately 4.5-fold greater potency than cucurbitacin C (CuC; $IC_{50} = 8.12 \pm 0.31\ \mu M$). Preliminary structure–activity relationship analysis suggested that the presence of a C-25 acetyl group and a side-chain $\alpha,\beta$-unsaturated ketone was associated with greater cytotoxic activity, whereas a free hydroxyl group at C-19 appeared unfavorable. These findings position CuC7 as a particularly promising member of this analogue series for further mechanistic investigation. Despite its potent cytotoxicity, the molecular targets and mechanisms underlying the cytotoxic effects of CuC7 in prostate cancer remain insufficiently characterized.

Integrated computational pharmacology provides a systematic strategy for investigating potential molecular targets and mechanism-of-action pathways of bioactive compounds with poorly characterized pharmacological profiles (Kumar and Patil, 2025). To address this gap, the present study integrated network pharmacology, molecular docking, 400-ns molecular dynamics simulations, MM/PBSA calculations, and cell-based assays to investigate CuC7 in a prostate cancer context. Accordingly, this study seeks to relate the cytotoxic phenotype of CuC7 to the molecular biology of prostate cancer by identifying and experimentally evaluating plausible target–pathway relationships. The overall workflow is presented in Fig. 1. The study also highlights the potential of underutilized cucumber biomass as a source of bioactive compounds for drug-discovery research.

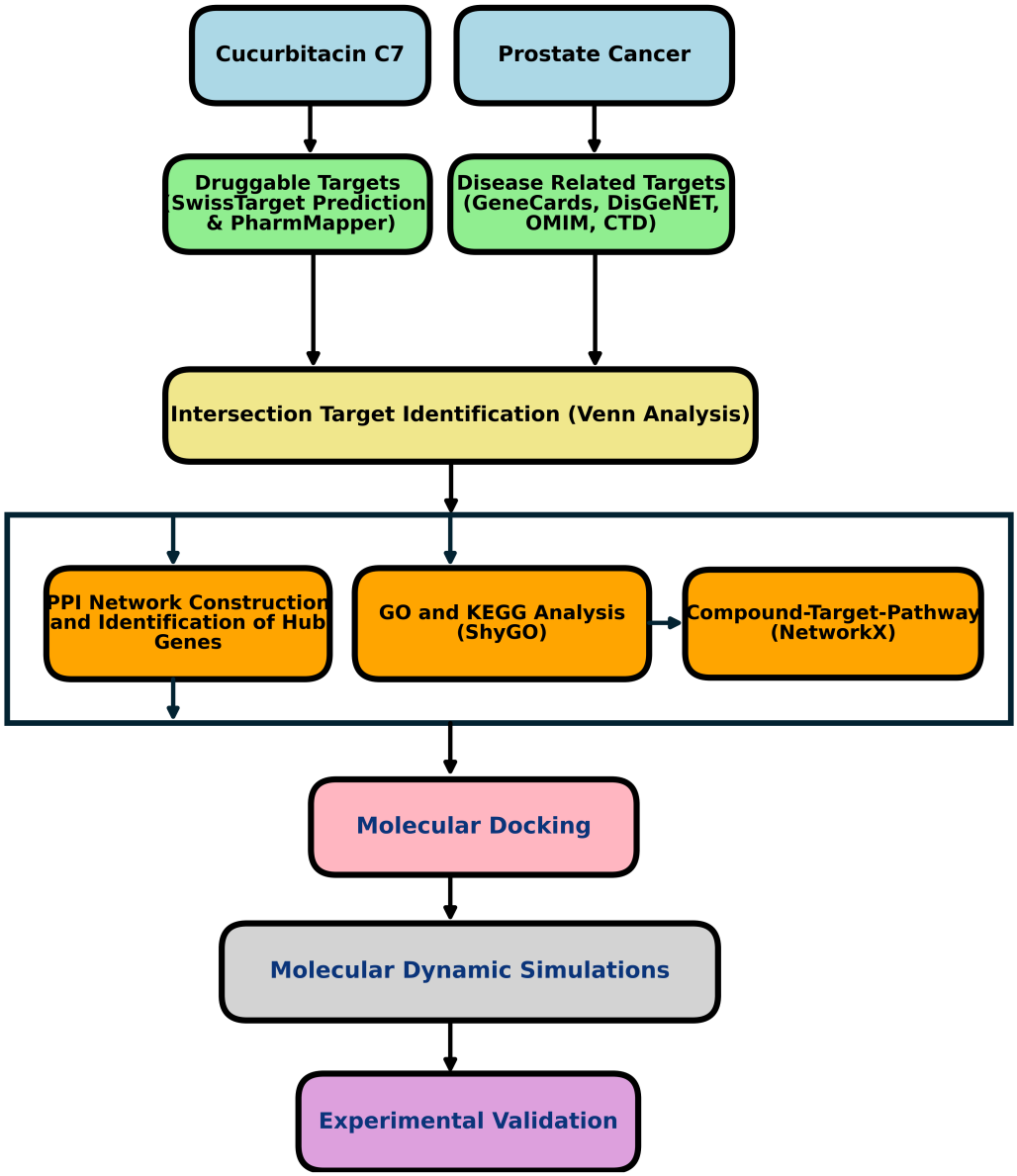



**Figure 1:** Overall workflow integrating network pharmacology, molecular docking, molecular dynamics simulations, and experimental validation to evaluate the therapeutic potential of Cucurbitacin C7.

# 2 Methodology

## 2.1 Network pharmacology analysis

The structure of CuC7 was obtained from Qing et al. (2022), reconstructed in Avogadro (Hanwell et al., 2012), and converted to a canonical simplified molecular-input line-entry system (SMILES) representation. The SMILES string was submitted to SwissTargetPrediction (http://www.swisstargetprediction.ch/; accessed 10 December 2025), and targets with a predicted probability

of at least 0.1 were retained. Complementary predictions were obtained from PharmMapper (http://www.lilab-ecust.cn/pharmmapper/; accessed 10 December 2025), for which hits with a normalized fit score of at least 0.6 were retained (Liu et al., 2010). Target identifiers were standardized to human UniProt gene symbols, the two result sets were merged, and duplicates were removed. The two platforms were combined to broaden target coverage and reduce dependence on a single prediction algorithm; this approach does not eliminate algorithmic or database bias.

PCa-associated genes were collected from GeneCards (https://www.genecards.org/; accessed 10 December 2025), DisGeNET v25.3 (https://www.disgenet.org/; accessed 10 December 2025), Online Mendelian Inheritance in Man (OMIM; https://www.omim.org/; accessed 10 December 2025), and the Comparative Toxicogenomics Database (CTD; https://ctdbase.org/; accessed 11 December 2025) (Stelzer et al., 2016). A GeneCards relevance score greater than 10 and a CTD inference score of at least 20 were required, whereas only curated associations were retained from DisGeNET and OMIM. Records were standardized to human gene symbols, merged, and deduplicated. The intersection between the resulting disease-gene set and the predicted CuC7 target set was defined as the candidate target set.

A PPI network was obtained using STRING v12.0 (Szklarczyk et al., 2021). The network was generated for *Homo sapiens* using the full STRING network, with evidence-based associations filtered at a minimum combined interaction score of 0.400 (medium confidence). Evidence from text mining, experiments, databases, co-expression, neighborhood, gene fusion, and co-occurrence was considered, with no additional first- or second-shell interactors included. All query proteins were retained in the network. The network was visualized in Cytoscape 3.10.1 (Shannon et al., 2003). Node topology was analyzed with CytoNCA (Tang et al., 2015) using degree, betweenness, closeness, eigenvector centrality, local average connectivity, and network centrality. These measures were selected to capture complementary aspects of connectivity, brokerage, proximity, influence, and local and global network organization. Hub proteins were defined as the intersection of the ten highest-ranked nodes under all six measures.

Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses were conducted in ShinyGO 0.80 (Ge et al., 2020) using *Homo sapiens*, with all human genes as the background gene set. An FDR threshold of 0.05 was applied using the Benjamini–Hochberg multiple-testing correction. The 10 terms with the lowest FDR values within each GO category and the 15 KEGG pathways with the lowest FDR values were retained for visualization. A compound–target–pathway network was constructed using NetworkX 3.5 in Python 3.14.0. Hub-target occurrence percentages were calculated by dividing the number of selected GO terms and KEGG pathways containing each hub protein by the total number of selected terms and pathways and multiplying by 100.

## 2.2 Molecular docking

Protein coordinates were obtained from the Protein Data Bank (PDB) (Bittrich et al., 2023). Redocking validation was performed for AKT1 (PDB 3O96), MDM2 (PDB 4HG7), PTGS2 (PDB 3LN1), EGFR (PDB 1M17), PIK3CA (PDB 7PG6, chain A; alpelisib/NVP-BYL719, ligand code 1LT), ERBB2 (PDB 3PP0, chain A; SYR127063, ligand code 03Q), and BCL2L1 (PDB 2YXJ, chain A; ABT-737, ligand code N3C). AKT1, MDM2, PTGS2, and EGFR were subsequently selected for MD simulations. For these four receptors, chain A was used and missing atoms and residues were repaired before docking. Alternate-location records were inspected; conformer A was retained when

resolving alternate conformers in MDM2 and EGFR, whereas the deposited alternate-conformer records were retained for AKT1 and PTGS2. All crystallographic waters were removed. Cofactor and heteroatom records were also inspected: the sulfate record in MDM2 was identified, HEM was retained in PTGS2 and its Fe field was repaired, and no cofactors were included for AKT1 or EGFR. PDB 3LN1 is a murine PTGS2 structure and was selected because it contains co-crystallized celecoxib; the species difference is considered when interpreting the PTGS2 results.

Co-crystallized ligands were removed for CuC7 docking. Polar hydrogen atoms and Kollman charges were added to the receptors using AutoDockTools 1.5.7 (Morris et al., 2009), followed by conversion to PDBQT format. CuC7 was constructed in Avogadro and minimized for 200 iterations with the MMFF94 force field. Gasteiger partial charges were assigned to CuC7 and the reference ligands before conversion to PDBQT format.

Docking was performed with AutoDock Vina 1.2.7 using the Vina scoring function (Eberhardt et al., 2021). Nine binding modes were generated with an exhaustiveness of 8, an energy range of 4 kcal/mol, and eight CPU threads. For all seven targets, the docking box was centered on the corresponding co-crystallized ligand and extended by 5 Å in all directions; the resulting coordinates are reported in Table 1. Poses were ranked by the Vina score and inspected for interactions within the crystallographic binding site. Protein–ligand interactions were visualized using BIOVIA Discovery Studio 2025.

The co-crystallized inhibitors were prepared with the same ligand workflow and redocked into their respective structures. Heavy-atom RMSDs between crystallographic and redocked ligand poses were calculated using RDKit 2025.09.1 following receptor-structure alignment. Atom correspondence was established by maximum-common-substructure (MCS) mapping of chemically equivalent ligand atoms. RMSDs were calculated in the fixed receptor coordinate frame without independent superposition of the ligand poses, as defined in Eq. 1.

$$\mathrm{RMSD} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}[(x_i - x_i')^2 + (y_i - y_i')^2 + (z_i - z_i')^2]}. \tag{1}$$

Here, $N$ is the number of matched heavy atoms, and $(x_i, y_i, z_i)$ and $(x_i', y_i', z_i')$ denote the Cartesian coordinates of corresponding atoms in the crystallographic and redocked ligand poses, respectively. Reference-ligand identities, docking boxes, docking scores, and redocking RMSDs are reported in Tables 1 and 2.

**Table 1:** Structures and grid parameters used for reference-ligand redocking and CuC7 docking of all seven targets. Grid coordinates are in Å.

| Target (PDB) | Reference ligand (code) | Grid center $(x, y, z)$ | Grid size $(x, y, z)$ | Padding | Exhaustiveness |
|---|---|---|---|---|---|
| AKT1 (3O96) | AKT inhibitor VIII (IQO) | 9.657, −7.762, 10.604 | 17.712, 25.771, 22.332 | 5 Å | 8 |
| MDM2 (4HG7) | Nutlin-3a (NUT) | −23.945, 8.091, −13.667 | 19.836, 22.538, 18.581 | 5 Å | 8 |
| PTGS2 (3LN1) | Celecoxib (CEL) | 30.597, −22.559, −15.758 | 20.438, 15.462, 18.904 | 5 Å | 8 |
| EGFR (1M17) | Erlotinib (AQ4) | 21.698, 0.303, 52.092 | 25.709, 14.684, 17.099 | 5 Å | 8 |
| PIK3CA (7PG6) | Alpelisib/NVP-BYL719 (1LT) | −1.249, −9.012, 17.457 | 16.805, 18.038, 22.972 | 5 Å | 8 |
| ERBB2 (3PP0) | SYR127063 (03Q) | 17.099, 16.549, 26.601 | 17.372, 23.936, 17.250 | 5 Å | 8 |
| BCL2L1 (2YXJ) | ABT-737 (N3C) | −9.874, −14.522, 10.799 | 23.393, 29.337, 16.278 | 5 Å | 8 |

## 2.3 Molecular dynamics simulations

For each principal target, apo, CuC7-bound, and reference-inhibitor-bound systems were prepared and simulated using GROMACS 2023.5 (Abraham et al., 2015). Proteins were described with the CHARMM36m force field (Huang et al., 2017), while CuC7 and the reference inhibitors were parameterized using CGenFF program version 2.5 with CGenFF force-field version 4.6. Protein and ligand topologies were generated using compatible CHARMM parameters and integrated into the complete simulation systems. CGenFF parameter quality was assessed from the assigned penalty scores; the maximum charge and bonded-parameter penalties were both below 10. A penalty score of 10 was used as the threshold for additional validation, whereas scores above 50 would require parameter optimization. Consequently, no additional parameter optimization was required.

Each solute was placed in a cubic periodic box with a minimum solute-to-box distance of 1.0 nm and solvated with TIP3P water. Counterions were added to neutralize the system, followed by NaCl to a final concentration of 0.15 M. Energy minimization used the steepest-descent algorithm for a maximum of 50,000 steps with an energy-convergence criterion of 1000 kJ $\mathrm{mol}^{-1}$ $\mathrm{nm}^{-1}$. Systems were equilibrated for 100 ps under constant volume and temperature (NVT) conditions at 310 K and for 100 ps under constant pressure and temperature (NPT) conditions at 310 K and 1 bar. Initial velocities were drawn from a Maxwell–Boltzmann distribution at 310 K. Production simulations were propagated for 400 ns under NPT conditions at 310 K and 1 bar. Temperature and pressure were maintained using the Berendsen thermostat and barostat, respectively, with coupling constants of 10 ps for both methods. A 2-fs integration time step and three-dimensional periodic boundary conditions were used. Short-range electrostatic and van der Waals interactions were truncated at 1.2 nm. Long-range electrostatics were treated with particle mesh Ewald, using a grid spacing of 0.12 nm and fourth-order interpolation. A Verlet neighbor list updated every 10 steps was used, and bonds involving hydrogen atoms were constrained using LINCS (Hess et al., 1997). Position restraints were applied during NVT and NPT equilibration and removed during production. Coordinates were recorded every 10 ps. One independent 400-ns production trajectory was generated for each simulated system.

Initial velocities were generated using automatically selected pseudorandom seeds (`gen_seed = -1`).

Backbone RMSD, per-residue RMSF, radius of gyration ($R_g$), and protein–ligand hydrogen bonds were calculated from the production trajectories. Apo and reference-inhibitor trajectories were analyzed with the corresponding protocol to provide within-target context. Before analysis, trajectories were sequentially made whole and centered using the `-pbc mol` and `-center` options, followed by removal of periodic-boundary discontinuities using `-pbc nojump`. RMSD was calculated for protein-backbone atoms after fitting each frame to the protein backbone. RMSF was calculated for C$\alpha$ atoms after least-squares fitting using the protein backbone. Protein–ligand hydrogen bonds were defined using a donor–acceptor distance of $\leq 3.5$ Å and a donor–hydrogen–acceptor angle of $\geq 120°$. The complete 400-ns trajectories were displayed, while the final 100 ns were used for the primary quantitative analyses. The reported curves and averages were obtained from the single trajectory generated for each system. No formal numerical convergence threshold was prespecified; operational structural stability was assessed from plateau behavior and the absence of sustained drift in backbone RMSD and radius of gyration during the final 100 ns.

## 2.4 MM/PBSA calculations

Binding energies were estimated with the molecular mechanics/Poisson–Boltzmann surface area (MM/PBSA) approach using gmx_MMPBSA 1.6.2 with its built-in PBSA solver (Valdés-Tresanco et al., 2021). For each protein–ligand system,

$$\Delta G_{\mathrm{bind}} = G_{\mathrm{complex}} - (G_{\mathrm{protein}} + G_{\mathrm{ligand}}), \tag{2}$$

and the reported estimate was decomposed as

$$\Delta G_{\mathrm{bind}} = \Delta E_{\mathrm{vdW}} + \Delta E_{\mathrm{elec}} + \Delta G_{\mathrm{polar}} + \Delta G_{\mathrm{SASA}}, \tag{3}$$

where $\Delta E_{\mathrm{vdW}}$ and $\Delta E_{\mathrm{elec}}$ are the van der Waals and electrostatic terms, $\Delta G_{\mathrm{polar}}$ is the polar solvation term, and $\Delta G_{\mathrm{SASA}}$ is the non-polar solvent-accessible-surface-area term.

The Poisson–Boltzmann and non-polar models were specified using `ipb = 2` and `inp = 1`, respectively. The internal and external dielectric constants were 1.0 and 80.0, respectively, with an ionic strength of 0.150 M. CHARMM atomic radii were used (`PBRadii = 7`; `radiopt = 0`). Solvent-accessible-surface-area calculations used `sasopt = 0`, a solvent-probe radius of 1.4 Å, a mobile-ion probe radius of 2.0 Å, and an arc resolution of 0.25 Å. The Poisson–Boltzmann grid scale was 2.0 $\mathrm{Å}^{-1}$, corresponding to a grid spacing of 0.50 Å.

Calculations used 1,001 snapshots extracted every 100 ps from the final 100 ns (300–400 ns) of each 400-ns production trajectory. Values are reported as the mean ± standard deviation across sampled frames. Entropic contributions were not calculated. All reference-inhibitor values and uncertainties were calculated from the corresponding trajectories and verified against the raw gmx_MMPBSA outputs.

## 2.5 Experimental assessment

CuC7 was isolated from the leaves of cucumber (*Cucumis sativus* L.) line 9930 following the procedure described by Qing et al. (2022a). The isolated compound was characterized, and its purity was determined to be 95.8% by high-performance liquid chromatography (HPLC) before use in the subsequent cell-based assays.

DU145 human prostate cancer cells (ATCC, HTB-81) were maintained in Dulbecco's modified Eagle medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and 1% penicillin–streptomycin at 37 °C in a humidified atmosphere containing 5% $CO_2$. Cells tested negative for mycoplasma contamination before experimentation. Cells at passage numbers no greater than 15 were used for all experiments. Cell-line identity was confirmed by short tandem repeat (STR) profiling. Cells were seeded in 96-well plates at $5 \times 10^3$ cells per well and allowed to adhere overnight. **[AUTHOR QUERY:** Report the cell-line authentication date.] CuC7 was dissolved in dimethyl sulfoxide (DMSO) and diluted in complete medium. The final DMSO concentration was maintained below 0.1% (v/v). The vehicle control contained the same final DMSO concentration as the CuC7-treated groups. Cells were treated for 24 h with 0.10, 0.25, 0.50, 0.75, 1.0, 2.5, 5.0, 7.5, or 10.0 $\mu$M CuC7. The CCK-8 experiment was performed as three independent biological replicates, with technical replicate wells for each treatment in each experiment.

Cell viability was assessed using Cell Counting Kit-8 (CCK-8; Dojindo Molecular Technologies, Inc., Kumamoto, Japan; product code CK04). A 10 $\mu$L aliquot of CCK-8 reagent was added per well,

and the plates were incubated for 2 h at 37 °C before absorbance was measured at 450 nm using a BioTek ELx808 microplate reader. Concentration–response data were fitted by nonlinear regression to a four-parameter logistic (4PL) model using GraphPad Prism 10.2, as defined in Eq. 4.

$$Y = \text{Bottom} + \frac{\text{Top} - \text{Bottom}}{1 + 10^{(\text{LogIC}_{50} - X) \times \text{HillSlope}}}. \tag{4}$$

Here, $Y$ is the predicted response, $X$ is the base-10 logarithm of the CuC7 concentration, Top and Bottom are the response plateaus, $\text{LogIC}_{50}$ is the base-10 logarithm of the half-maximal inhibitory concentration, and HillSlope describes the curve steepness. A 95% confidence interval was calculated for the $\text{IC}_{50}$ estimate. ANOVA assumptions were assessed using residual diagnostics for normality and homogeneity of variance. Group differences in cell viability were assessed using one-way analysis of variance (ANOVA), followed by Tukey's multiple-comparisons test, with $p < 0.05$ considered statistically significant.

AKT1 protein abundance was measured by enzyme-linked immunosorbent assay (ELISA) after cells were treated with 10 $\mu$M CuC7 for 24 h. Cells were lysed in ice-cold RIPA buffer containing 50 mM Tris–HCl, 150 mM NaCl, 1% NP-40, 0.5% sodium deoxycholate, and 0.1% SDS, supplemented with protease inhibitors. Lysates were incubated on ice for 5 min, sonicated, and centrifuged at 14,000 rpm for 10 min at 4 °C. Total protein was quantified by BCA assay following incubation at 37 °C for 30 min, and ELISA-derived AKT1 concentrations were normalized to total protein. The Human AKT1 (Total) Coated ELISA Kit (Invitrogen, Thermo Fisher Scientific; catalog no. KHO0531) was used according to the manufacturer's instructions. Absorbance was recorded at 450 nm and AKT1 concentrations were normalized to the control. ELISA measurements were obtained from three technical replicate wells per group in a single experiment and were analyzed using GraphPad Prism 10.2. Data are presented as mean ± standard deviation (SD) and were summarized descriptively without inferential statistical testing. Percentage reduction in AKT1 abundance was calculated as

$$\%\,\text{reduction} = \frac{C_{\text{control}} - C_{\text{CuC7}}}{C_{\text{control}}} \times 100, \tag{5}$$

where $C_{\text{control}}$ and $C_{\text{CuC7}}$ are the total-protein-normalized AKT1 concentrations in the control and CuC7-treated groups, respectively.

# 3 Results

## 3.1 Candidate targets and hub-protein prioritization

Target prediction produced 362 non-redundant putative CuC7-associated proteins, while the integrated disease databases produced 4,722 PCa-associated genes. Their intersection contained 32 candidate proteins (Fig. 2A). The resulting STRING network included all 32 candidate proteins, with no disconnected nodes, and showed extensive connectivity among them (Fig. 2B). The intersection of the ten highest-ranked nodes under all six centrality measures identified seven hub proteins: MDM2, PIK3CA, AKT1, EGFR, PTGS2, BCL2L1, and ERBB2 (Fig. 3). These nodes were prioritized for subsequent interpretation because they were consistently ranked across complementary network metrics; their centrality does not by itself demonstrate that they are bound or regulated by CuC7.

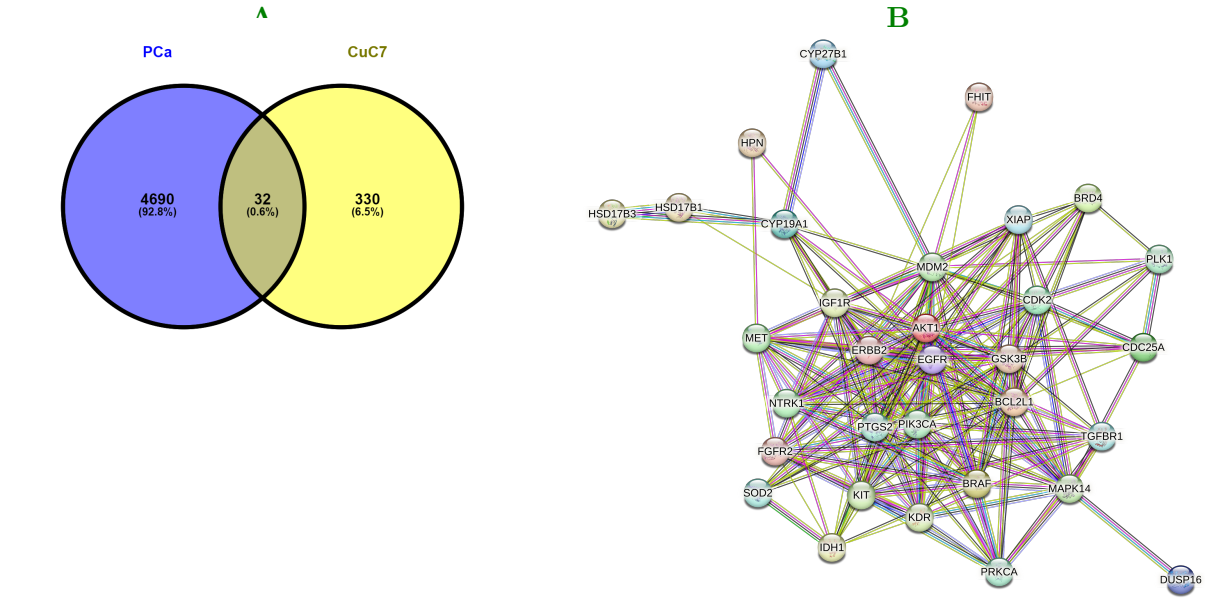



**Figure 2:** Candidate therapeutic targets of CuC7 in prostate cancer. (A) Venny intersection of 4,722 PCa-associated genes and 362 non-redundant predicted CuC7 targets. The regions contain 4,690 PCa-only genes, 32 shared targets, and 330 CuC7-only targets; percentages are calculated relative to the union of 5,052 unique genes. (B) STRING v12.0 protein–protein interaction network of all 32 shared targets. Edge colors represent the STRING evidence channels specified in Section 2.1; no additional interactors were included.

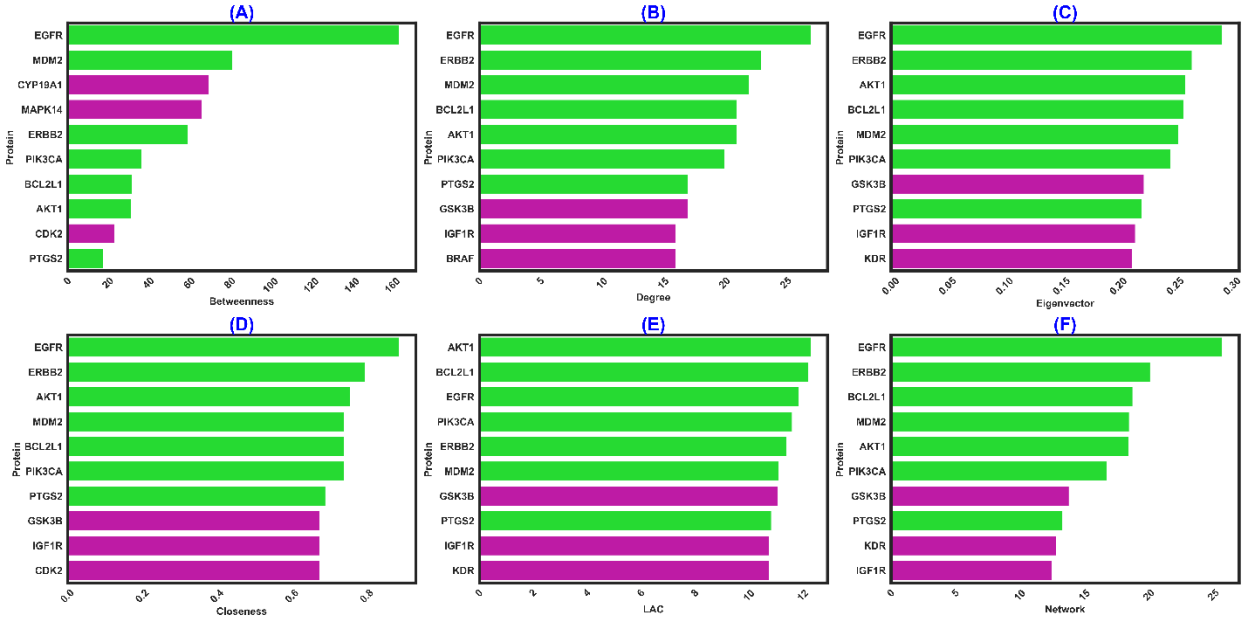



**Figure 3:** Top ten nodes ranked by (A) degree, (B) betweenness, (C) closeness, (D) eigenvector centrality, (E) local average connectivity, and (F) network centrality. Green bars denote proteins retained in the intersection across all six top-ten lists; magenta bars denote other highly ranked proteins.

## 3.2 Functional enrichment

GO analysis associated the 32 candidate targets with apoptosis-related processes, protein phosphorylation, intracellular signal transduction, kinase and transferase activity, receptor complexes, mitochondria, membrane rafts, and cell junctions (Fig. 4A). KEGG analysis identified enrichment in the PI3K–AKT, MAPK, FoxO, focal-adhesion, prostate-cancer, and EGFR tyrosine-kinase-inhibitor-resistance pathways (Fig. 4B).

Across the displayed terms, EGFR, AKT1, ERBB2, PIK3CA, MDM2, BCL2L1, and PTGS2 occurred at reported frequencies of 95.6%, 84.4%, 77.8%, 71.1%, 46.7%, 37.8%, and 28.9%, respectively (Fig. 5). The compound–target–pathway network connected CuC7 predictions to several signaling axes (Fig. 6). These results support prioritization of a multi-protein hypothesis but do not establish simultaneous biochemical modulation of the represented proteins.

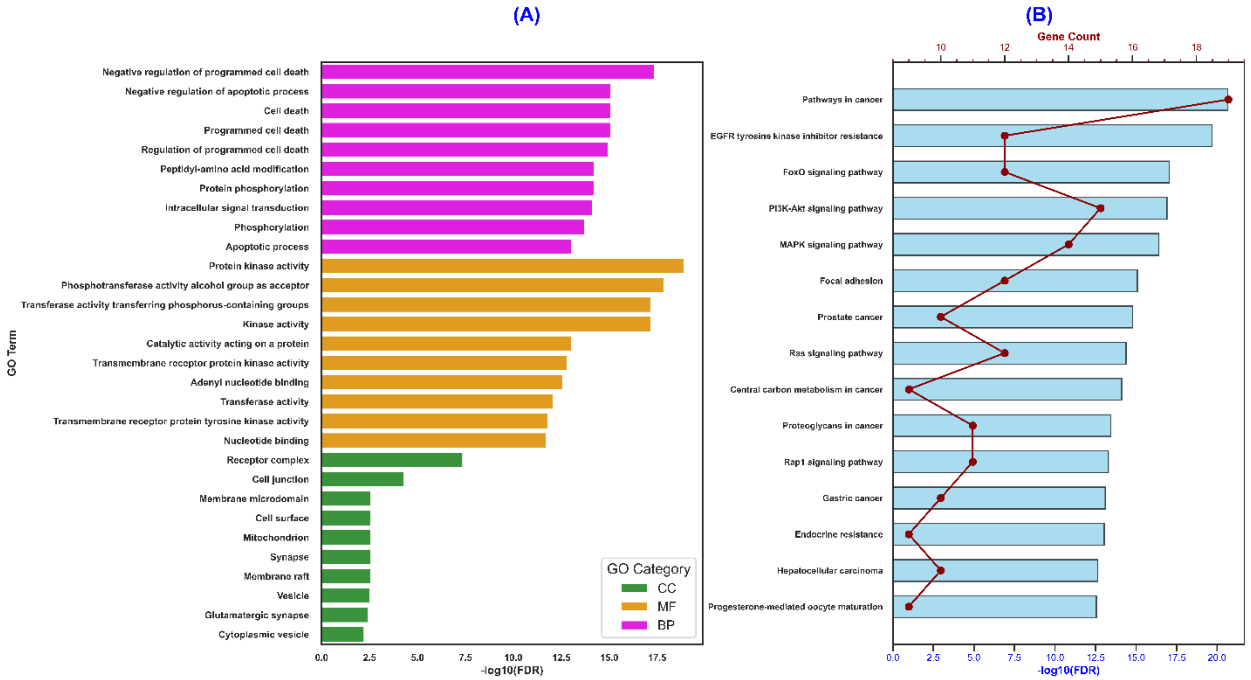



**Figure 4:** Functional enrichment of the 32 candidate targets. (A) Selected Gene Ontology biological-process, molecular-function, and cellular-component terms. Horizontal bars show $-\log_{10}(\text{FDR})$, and colors denote the three GO categories. (B) Selected KEGG pathways. Blue bars show $-\log_{10}(\text{FDR})$ on the lower x-axis, whereas the red line and points show the number of candidate genes assigned to each pathway on the upper x-axis. Enrichment analyses used all human genes as the background and a Benjamini–Hochberg FDR threshold of 0.05. The 10 terms with the lowest FDR values in each GO category and the 15 KEGG pathways with the lowest FDR values are displayed.

| Category | Term | MDM2 | PIK3CA | AKT1 | EGFR | PTGS2 | BCL2L1 | ERBB2 |
|---|---|---|---|---|---|---|---|---|
| BP | Negative regulation of programmed cell death | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| BP | Cell death | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| BP | Programmed cell death | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| BP | Negative regulation of apoptotic process | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| BP | Regulation of programmed cell death | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| BP | Protein phosphorylation | | ✓ | ✓ | ✓ | ✓ | | ✓ |
| BP | Peptidyl-amino acid modification | ✓ | | ✓ | ✓ | ✓ | | ✓ |
| BP | Intracellular signal transduction | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| BP | Phosphorylation | | ✓ | ✓ | ✓ | ✓ | | ✓ |
| BP | Apoptotic process | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| MF | Protein kinase activity | | ✓ | ✓ | ✓ | | | ✓ |
| MF | Phosphotransferase activity alcohol group as acceptor | | ✓ | ✓ | ✓ | | | ✓ |
| MF | Kinase activity | | ✓ | ✓ | ✓ | | | ✓ |
| MF | Transferase activity transferring phosphorus-containing groups | | ✓ | ✓ | ✓ | | | ✓ |
| MF | Catalytic activity acting on a protein | ✓ | ✓ | ✓ | ✓ | | | ✓ |
| MF | Transmembrane receptor protein kinase activity | | | | ✓ | | | ✓ |
| MF | Adenyl nucleotide binding | | ✓ | ✓ | ✓ | | | ✓ |
| MF | Transferase activity | ✓ | ✓ | ✓ | ✓ | | | ✓ |
| MF | Transmembrane receptor protein tyrosine kinase activity | | | | ✓ | | | ✓ |
| MF | Nucleotide binding | | ✓ | ✓ | ✓ | | | ✓ |
| CC | Receptor complex | | | | ✓ | | | ✓ |
| CC | Cell junction | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ |
| CC | Mitochondrion | | | ✓ | | | ✓ | |
| CC | Cell surface | | | | ✓ | | | |
| CC | Membrane raft | | | | ✓ | ✓ | | |
| CC | Synapse | ✓ | | ✓ | ✓ | | ✓ | ✓ |
| CC | Membrane microdomain | | | | ✓ | ✓ | | |
| CC | Vesicle | ✓ | | ✓ | ✓ | | ✓ | ✓ |
| CC | Glutamatergic synapse | ✓ | | ✓ | ✓ | | | |
| CC | Cytoplasmic vesicle | ✓ | | | ✓ | | ✓ | ✓ |
| KEGG | Pathways in cancer | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| KEGG | EGFR tyrosine kinase inhibitor resistance | | ✓ | ✓ | ✓ | | ✓ | ✓ |
| KEGG | FoxO signaling pathway | ✓ | ✓ | ✓ | ✓ | | | |
| KEGG | PI3K-Akt signaling pathway | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ |
| KEGG | MAPK signaling pathway | | | ✓ | ✓ | | | ✓ |
| KEGG | Focal adhesion | | ✓ | ✓ | ✓ | | | ✓ |
| KEGG | Prostate cancer | ✓ | ✓ | ✓ | ✓ | | | ✓ |
| KEGG | Ras signaling pathway | | ✓ | ✓ | ✓ | | ✓ | |
| KEGG | Central carbon metabolism in cancer | | ✓ | ✓ | ✓ | | | ✓ |
| KEGG | Proteoglycans in cancer | ✓ | ✓ | ✓ | ✓ | | | ✓ |
| KEGG | Rap1 signaling pathway | | ✓ | ✓ | ✓ | | | |
| KEGG | Gastric cancer | | ✓ | ✓ | ✓ | | | ✓ |
| KEGG | Endocrine resistance | ✓ | ✓ | ✓ | ✓ | | | ✓ |
| KEGG | Hepatocellular carcinoma | | ✓ | ✓ | ✓ | | ✓ | |
| KEGG | Progesterone-mediated oocyte maturation | | ✓ | ✓ | | | | |
| METRICS | Term Count | 21 | 32 | 38 | 43 | 13 | 17 | 35 |
| METRICS | % of Terms | 46.7% | 71.1% | 84.4% | 95.6% | 28.9% | 37.8% | 77.8% |

**Figure 5:** Occurrence of prioritized hub proteins across the selected GO terms and KEGG pathways. Green marks indicate that a protein was assigned to a displayed term. The bottom row summarizes the reported percentage of displayed terms containing each protein. [**AUTHOR QUERY:** Confirm this interpretation of the percentages and define all color and symbol encodings.]

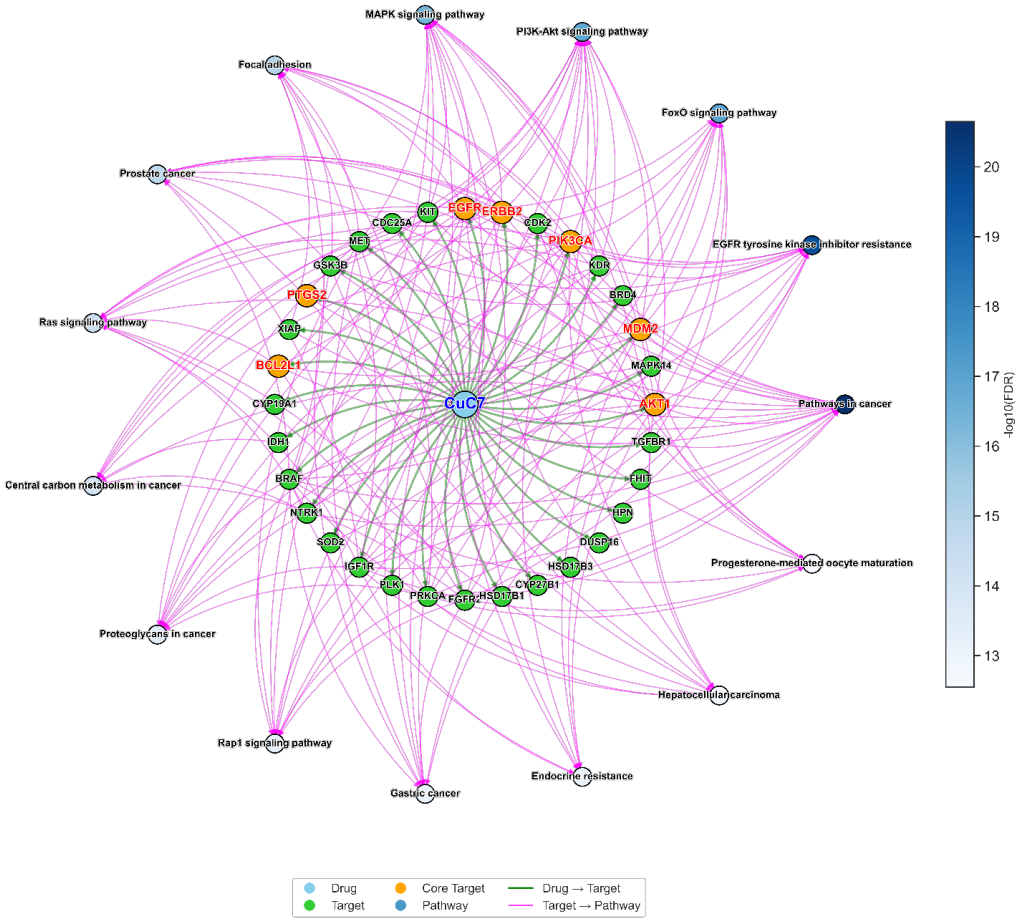



**Figure 6:** CuC7 candidate-target–pathway network. Node colors and shapes distinguish the compound, candidate proteins, and enriched pathways; edges denote predicted or database-derived associations rather than experimentally demonstrated CuC7–protein interactions.

## 3.3 Docking and crystallographic redocking

CuC7 was docked to AKT1, BCL2L1, EGFR, ERBB2, MDM2, PIK3CA, and PTGS2. The most negative CuC7 scores were obtained for PTGS2 (−8.73 kcal/mol), EGFR (−8.48 kcal/mol), MDM2 (−8.39 kcal/mol), and AKT1 (−8.18 kcal/mol), followed by ERBB2 (−7.06 kcal/mol), PIK3CA (−6.94 kcal/mol), and BCL2L1 (−6.75 kcal/mol). The selected CuC7 poses formed four predicted hydrogen bonds with AKT1, three with MDM2, and one each with EGFR and PTGS2, in addition to hydrophobic and van der Waals contacts (Fig. 7). AKT1, MDM2, PTGS2, and EGFR were selected for subsequent simulations on the basis of their docking scores and binding-site interaction profiles.

Redocking reproduced the co-crystallized ligand orientations with minimum heavy-atom RMSDs of 0.186 Å for AKT1, 0.534 Å for MDM2, 0.568 Å for PTGS2, 0.766 Å for PIK3CA, 0.979 Å for ERBB2, 1.320 Å for EGFR, and 1.490 Å for BCL2L1 (Table 2). The four targets selected for MD simulations are illustrated in Fig. 8. Under the same protocol, CuC7 had less negative docking

scores than the corresponding reference ligands for AKT1, PTGS2, PIK3CA, ERBB2, and BCL2L1 and more negative scores for MDM2 and EGFR. The comparisons provide protocol-matched context only; docking-score differences are not experimental affinity measurements.

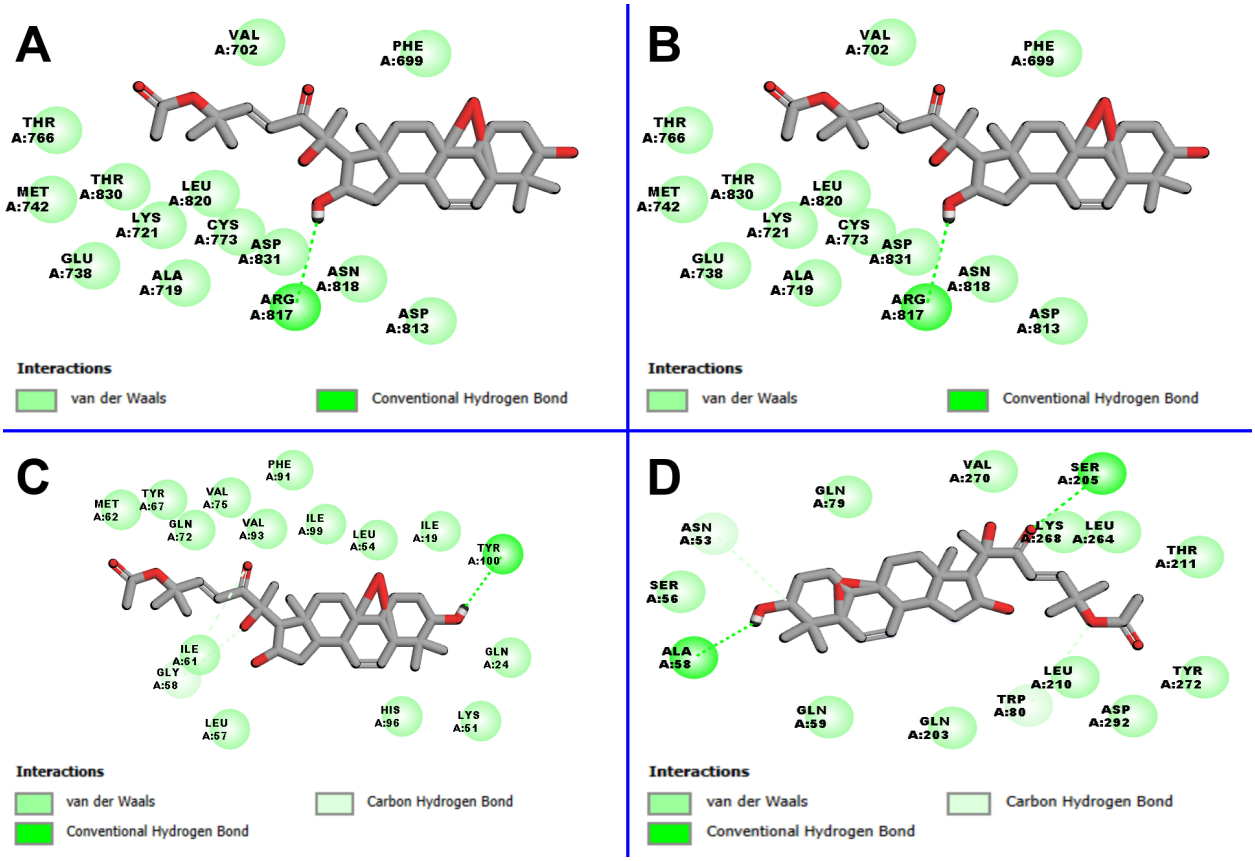



**Figure 7:** Two-dimensional interaction diagrams for selected CuC7 docking poses in (A) PTGS2, (B) EGFR, (C) MDM2, and (D) AKT1. Residue contacts and interaction types were assigned in BIOVIA Discovery Studio 2025. [**AUTHOR QUERY:** Panel A appears to duplicate the EGFR interaction diagram in panel B (including the same residue labels). Replace panel A with the verified PTGS2–CuC7 diagram and recheck every panel against its source complex.]

**Table 2:** Reference-ligand and CuC7 docking scores and minimum redocking RMSDs for all seven targets. Scores are Vina outputs and should not be interpreted as experimental binding free energies.

| Target (PDB) | Reference ligand | Reference score | CuC7 score | RMSD (Å) |
|---|---|---|---|---|
| AKT1 (3O96) | AKT inhibitor VIII | −14.420 | −8.18 | 0.186 |
| MDM2 (4HG7) | Nutlin-3a | −7.609 | −8.39 | 0.534 |
| PTGS2 (3LN1) | Celecoxib | −12.470 | −8.73 | 0.568 |
| EGFR (1M17) | Erlotinib (AQ4) | −7.055 | −8.48 | 1.320 |
| PIK3CA (7PG6) | Alpelisib/NVP-BYL719 (1LT) | −9.277 | −6.94 | 0.766 |
| ERBB2 (3PP0) | SYR127063 (03Q) | −10.870 | −7.06 | 0.979 |
| BCL2L1 (2YXJ) | ABT-737 (N3C) | −10.520 | −6.75 | 1.490 |

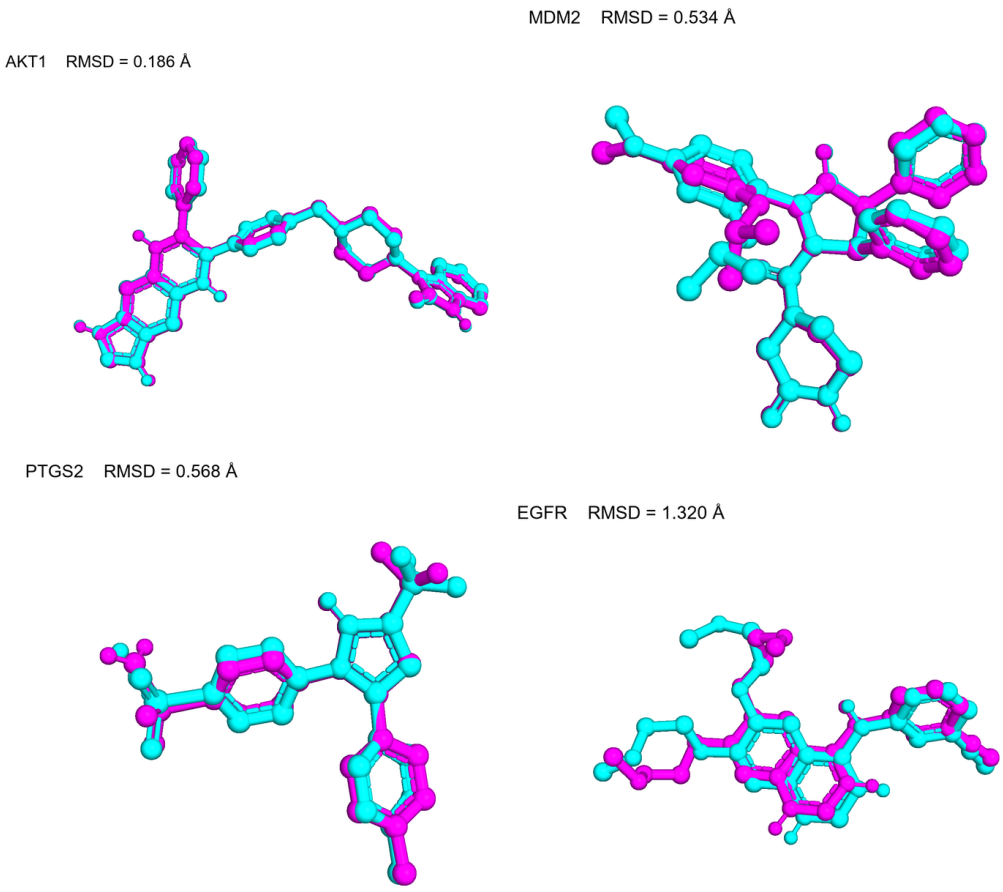



**Figure 8:** Crystallographic redocking validation for (A) AKT1, (B) MDM2, (C) PTGS2, and (D) EGFR. Experimental co-crystallized poses are shown in cyan and redocked poses in magenta. The minimum heavy-atom RMSDs were 0.186, 0.534, 0.568, and 1.320 Å, respectively.

## 3.4 Molecular dynamics simulations

### 3.4.1 Backbone RMSD

The CuC7-bound systems maintained comparatively stable backbone RMSD profiles over the 400-ns trajectories (Fig. 9). Mean RMSDs for the CuC7-bound systems were 1.97 Å for AKT1, 2.10 Å for EGFR, 2.37 Å for MDM2, and 2.39 Å for PTGS2. Reference-inhibitor and apo trajectories provide within-target context. These RMSD patterns describe structural deviation within the simulated systems and should not be interpreted as a ranking of experimental affinity across different proteins.

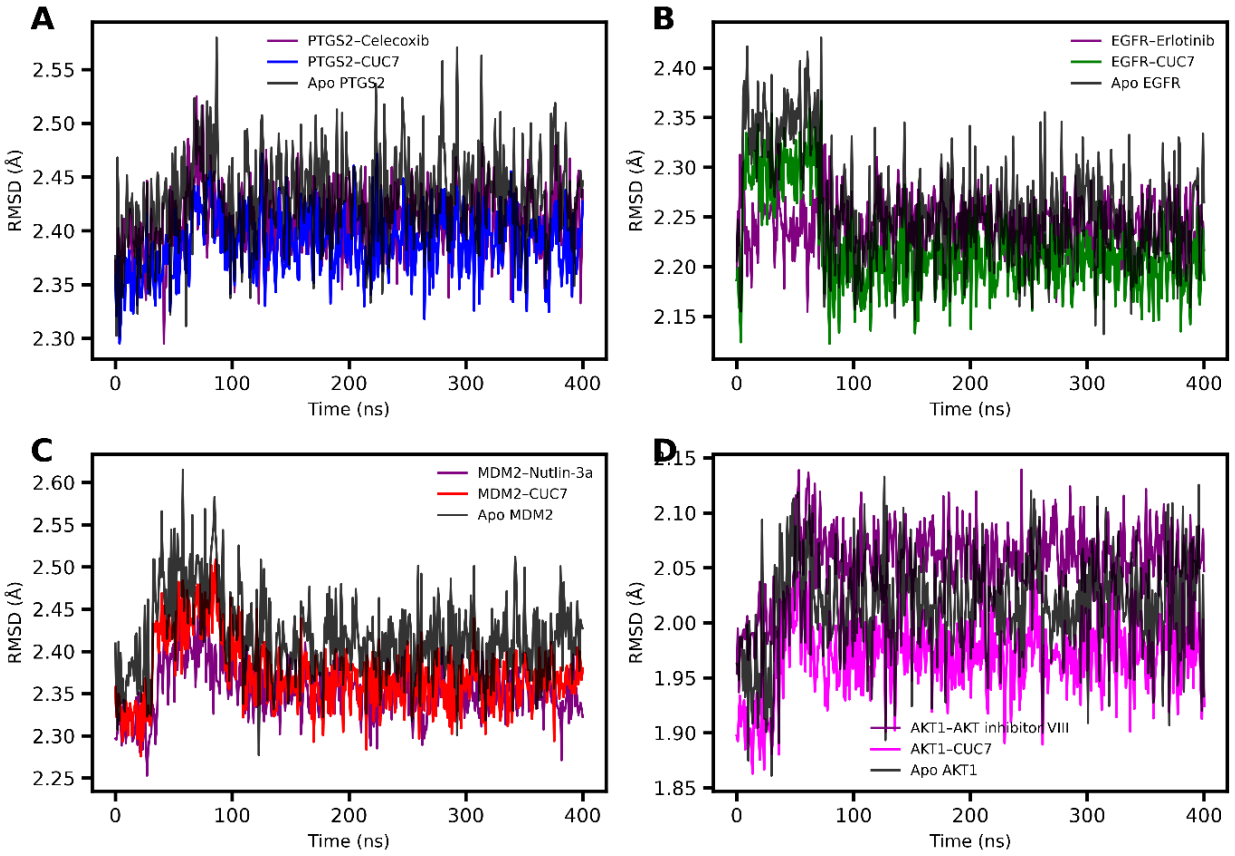



**Figure 9:** Backbone RMSD trajectories for apo, reference-inhibitor-bound, and CuC7-bound forms of (A) PTGS2, (B) EGFR, (C) MDM2, and (D) AKT1 over 400 ns of MD simulation.

### 3.4.2 Per-residue RMSF

RMSF analysis identified regions in which residue-level flexibility differed among apo, reference-inhibitor-bound, and CuC7-bound systems (Fig. 10). Mean RMSFs within the CuC7-bound trajectories were 0.69 Å for AKT1, 0.72 Å for PTGS2, 0.78 Å for MDM2, and 0.87 Å for EGFR. Local peaks remained in several regions. Because the proteins differ in length, fold, and residue composition, cross-protein differences in the mean RMSF were not treated as direct evidence of stronger binding.

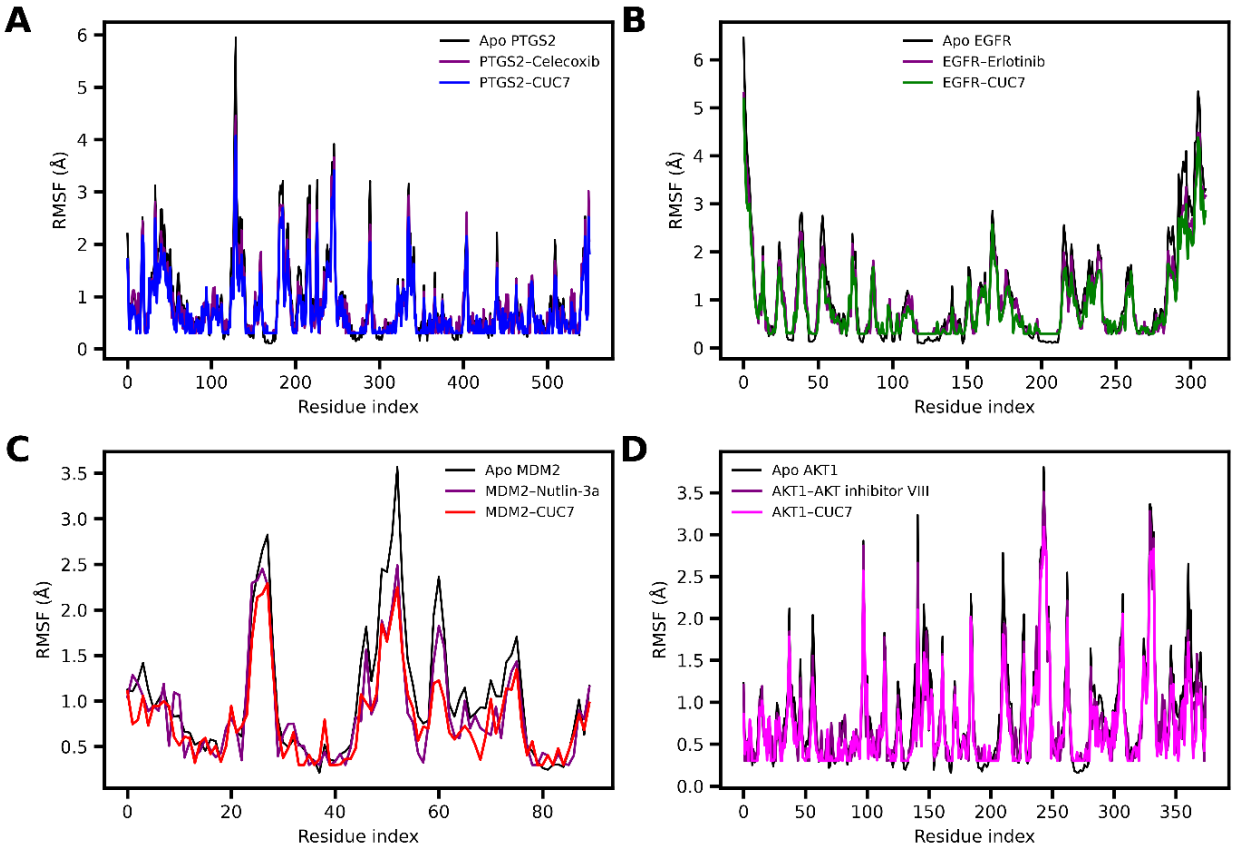



**Figure 10:** Per-residue RMSF profiles for apo, reference-inhibitor-bound, and CuC7-bound forms of (A) PTGS2, (B) EGFR, (C) MDM2, and (D) AKT1. Residue indices refer to the simulated constructs. [**AUTHOR QUERY:** Identify binding-site residues or domains, if analyzed, and report the fitted atom selection.]

### 3.4.3 Hydrogen bonds and radius of gyration

AKT1–CuC7 showed an average of 3.03 protein–ligand hydrogen bonds and a maximum of six. MDM2 and EGFR showed intermediate hydrogen-bond counts, while PTGS2 showed the lowest count among the four CuC7 systems (Fig. 11). The average pattern was AKT1 $>$ MDM2 $\approx$ EGFR $>$ PTGS2. Hydrogen-bond counts for the corresponding reference inhibitors were calculated with the same geometric definition.

The $R_g$ trajectories indicated differences in global compactness between apo and CuC7-bound systems (Fig. 12). The mean $R_g$ of the AKT1–CuC7 system was 20.09 Å. These global compactness measures were used as structural descriptors and not as direct measures of binding strength.

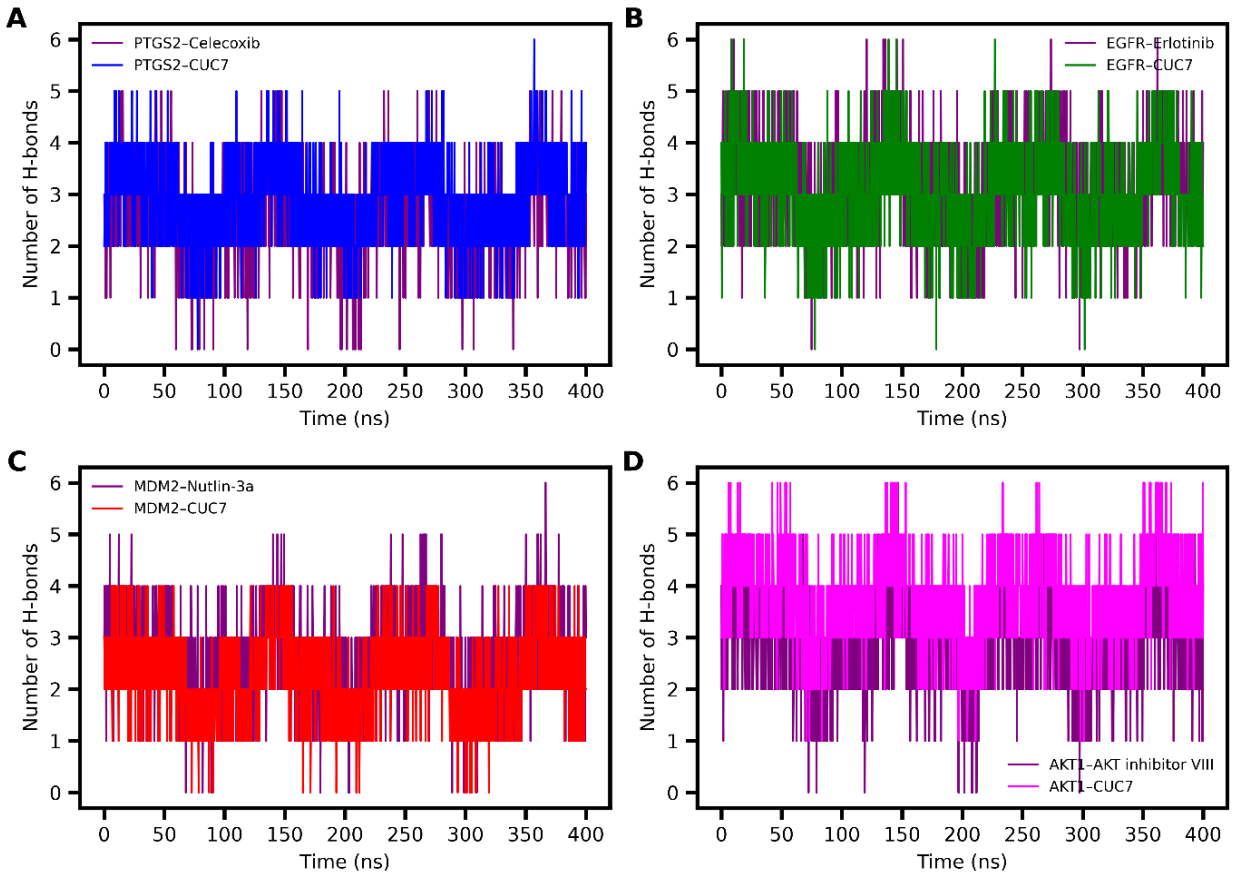



**Figure 11:** Protein–ligand hydrogen-bond counts for reference-inhibitor-bound and CuC7-bound (A) PTGS2, (B) EGFR, (C) MDM2, and (D) AKT1 systems over 400 ns. Hydrogen bonds were defined by a donor–acceptor distance of $\leq 3.5$ Å and a donor–hydrogen–acceptor angle of $\geq 120°$.

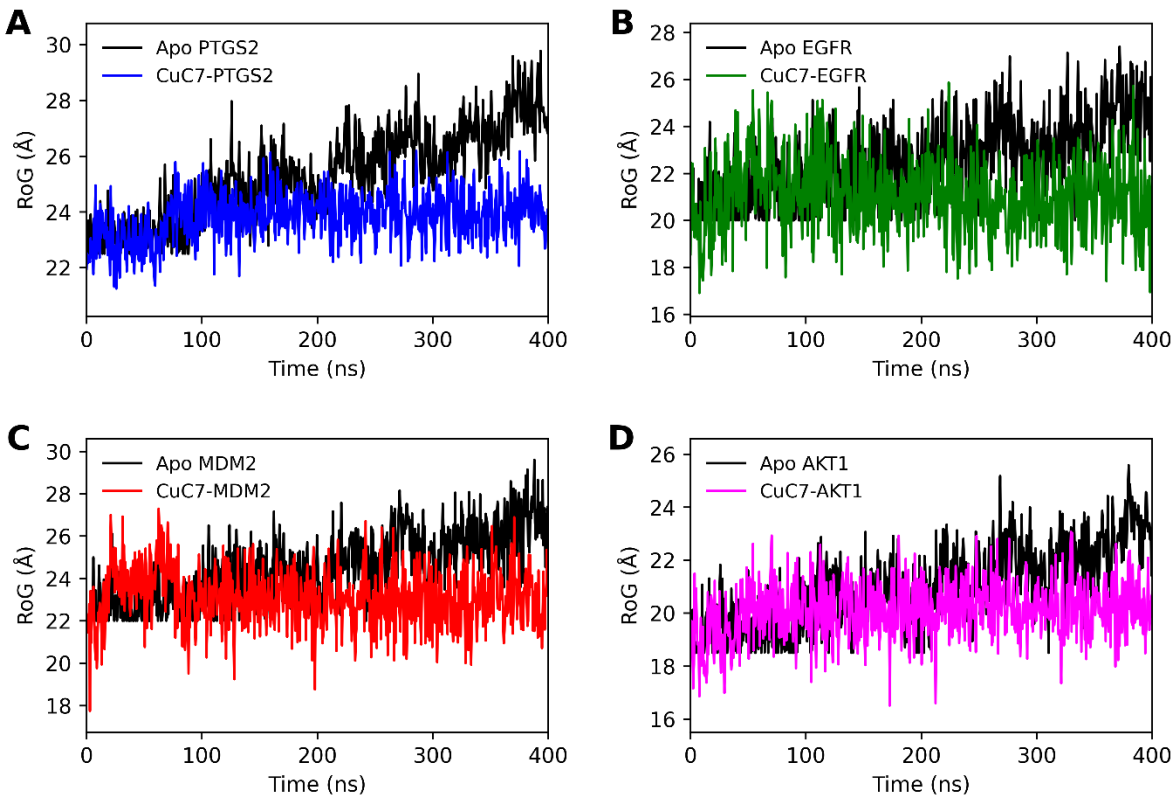



**Figure 12:** Radius-of-gyration trajectories for apo and CuC7-bound forms of (A) PTGS2, (B) EGFR, (C) MDM2, and (D) AKT1 over 400 ns.

## 3.5 Comparative MM/PBSA estimates

The mean MM/PBSA estimates for the CuC7 systems were $-59.36 \pm 3.00$ kcal/mol for AKT1, $-35.42 \pm 2.70$ kcal/mol for MDM2, $-20.58 \pm 2.70$ kcal/mol for PTGS2, and $-16.63 \pm 2.80$ kcal/mol for EGFR (Table 3; Fig. 13). Under the same analysis protocol, CuC7 produced more negative estimates than the corresponding reference inhibitors for AKT1, MDM2, and PTGS2, while the EGFR–CuC7 and EGFR–erlotinib estimates were similar. Van der Waals and electrostatic terms were favorable, whereas polar solvation opposed binding. These trajectory-derived values are approximate comparisons and do not confirm binding or quantify experimental affinity.

**Table 3:** MM/PBSA energy components for CuC7 and reference-inhibitor complexes (kcal/mol; mean ± SD across 1,001 sampled frames). Entropy was not included.

| System | $\Delta E_{\mathrm{vdW}}$ | $\Delta E_{\mathrm{elec}}$ | $\Delta G_{\mathrm{polar}}$ | $\Delta G_{\mathrm{SASA}}$ | $\Delta G_{\mathrm{bind}}$ |
|---|---|---|---|---|---|
| AKT1–AKT inhibitor VIII | $-34.00 \pm 3.80$ | $-28.00 \pm 3.40$ | $20.00 \pm 2.80$ | $-6.20 \pm 0.85$ | $-48.20 \pm 2.50$ |
| AKT1–CuC7 | $-38.79 \pm 4.30$ | $-36.48 \pm 4.00$ | $22.75 \pm 3.20$ | $-6.84 \pm 0.95$ | $-59.36 \pm 3.00$ |
| MDM2–Nutlin-3a | $-35.50 \pm 3.40$ | $-18.00 \pm 2.60$ | $28.00 \pm 3.20$ | $-5.80 \pm 0.80$ | $-31.30 \pm 2.30$ |
| MDM2–CuC7 | $-42.35 \pm 4.10$ | $-21.74 \pm 3.20$ | $35.37 \pm 4.10$ | $-6.70 \pm 0.90$ | $-35.42 \pm 2.70$ |
| PTGS2–celecoxib | $-31.50 \pm 3.20$ | $-16.80 \pm 2.80$ | $40.50 \pm 3.80$ | $-4.00 \pm 0.70$ | $-11.80 \pm 2.20$ |
| PTGS2–CuC7 | $-37.52 \pm 3.80$ | $-25.83 \pm 3.40$ | $46.48 \pm 4.50$ | $-3.72 \pm 0.75$ | $-20.58 \pm 2.70$ |
| EGFR–erlotinib | $-36.00 \pm 3.50$ | $-21.00 \pm 3.00$ | $45.00 \pm 4.20$ | $-4.20 \pm 0.75$ | $-16.20 \pm 2.40$ |
| EGFR–CuC7 | $-41.17 \pm 4.00$ | $-26.40 \pm 3.60$ | $55.79 \pm 5.00$ | $-4.85 \pm 0.85$ | $-16.63 \pm 2.80$ |

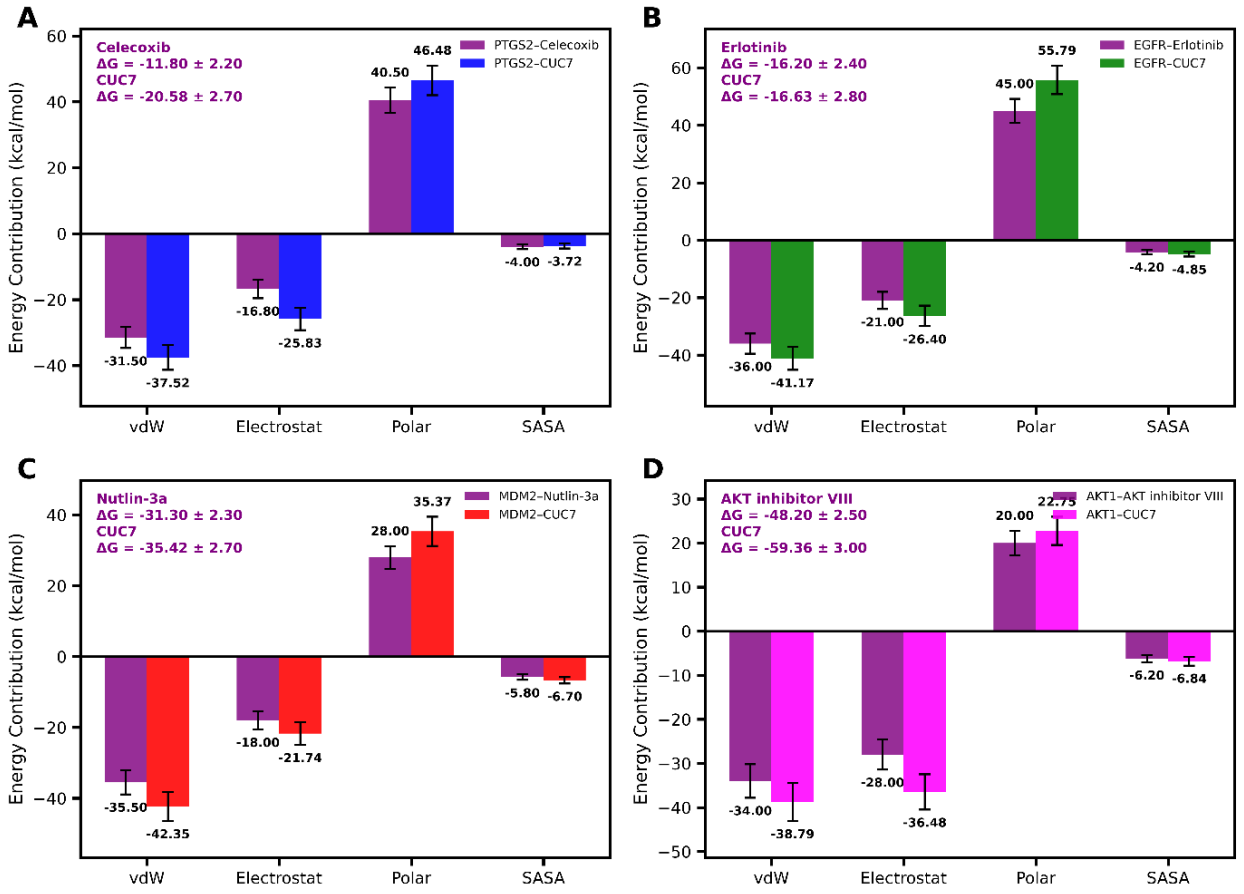



**Figure 13:** Comparative MM/PBSA energy components for CuC7 and the corresponding reference inhibitor in (A) PTGS2, (B) EGFR, (C) MDM2, and (D) AKT1 systems. Bars show means and error bars show standard deviations across 1,001 frames sampled from 300–400 ns.

## 3.6 Effects of CuC7 in DU145 cells

CuC7 reduced DU145 cell viability in a concentration-dependent manner after 24 h (Fig. 14A). One-way ANOVA showed a significant overall treatment effect, $F(9, 20) = 753.68$, $p < 0.001$. Tukey-

adjusted $p$ values versus the vehicle control were 1.000, 0.9995, and 0.0302 at 0.10, 0.25, and 0.50 $\mu$M, respectively; all comparisons at 0.75–10.0 $\mu$M had $p < 0.001$. Nonlinear regression yielded an $IC_{50}$ of 1.787 $\mu$M (95% CI, 1.654–1.928 $\mu$M). In the single ELISA experiment, total-protein-normalized AKT1 abundance was lower in lysates from cells treated with 10 $\mu$M CuC7 than in control lysates (Fig. 14B); the corresponding percentage reduction relative to the control is shown in Fig. 14C. This descriptive result indicates an associated change in AKT1 protein abundance but requires confirmation in independent biological experiments. These results demonstrate cytotoxic activity and an associated change in AKT1 protein abundance. They do not demonstrate direct CuC7–AKT1 binding, cellular target engagement, or inhibition of AKT1 kinase activity.

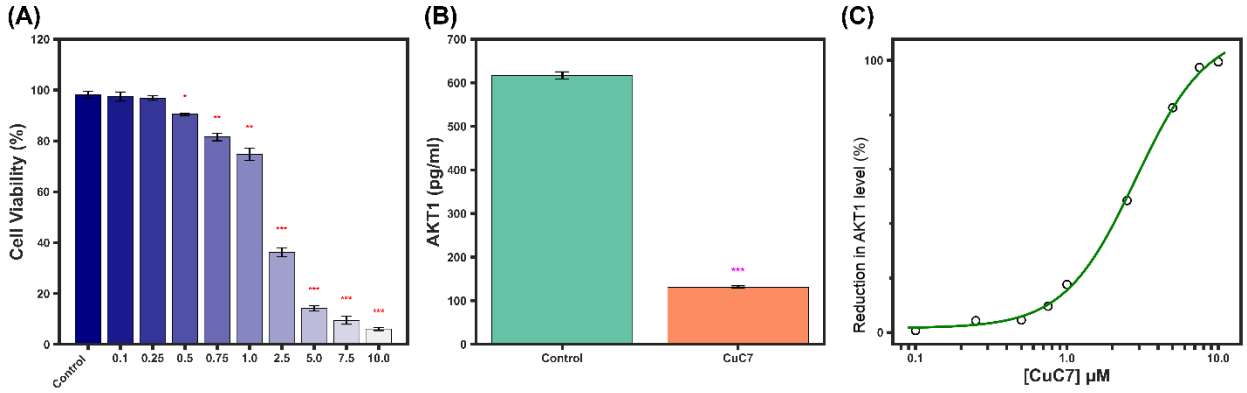



**Figure 14:** In vitro effects of CuC7 in DU145 cells after 24 h. (A) Cell viability measured by CCK-8 across the indicated CuC7 concentrations. (B) AKT1 protein concentration in control and 10 $\mu$M CuC7-treated lysates. (C) Percentage reduction in the total-protein-normalized AKT1 concentration at 10 $\mu$M CuC7 relative to the control, calculated using Eq. 5. ELISA data are shown as mean ± SD from three technical replicate wells per group in a single experiment. [**AUTHOR QUERY:** Define the statistical symbols and exact comparison used for each significance marker in panel A, and remove any inferential-statistics markers from panels B and C.]

# 4 Discussion

This study integrates target prediction, network analysis, docking validation, comparative simulations, energy estimation, and cell-based measurements to develop a testable hypothesis for CuC7 activity in prostate cancer. Thirty-two predicted targets overlapped with prostate cancer-associated genes, and seven proteins were consistently prioritized by six network metrics. The enrichment pattern connected these proteins to PI3K–AKT, MAPK, FoxO, receptor-tyrosine-kinase, apoptosis-related, and inflammatory pathways. These findings identify biologically relevant candidate relationships but remain dependent on database coverage and prediction algorithms.

The conceptual contribution extends earlier work that established the isolation, structure–activity relationships, and cytotoxicity of cucumber-derived cucurbitacins (Qing et al., 2022). Here, prostate cancer-specific network analysis was linked to crystallographic redocking, reference-inhibitor comparisons, 400-ns trajectories, MM/PBSA estimates, and DU145 measurements. This multi-layer design provides a more structured basis for prioritizing follow-up experiments than cytotoxicity or unvalidated docking alone. It nevertheless remains a hypothesis-generation framework rather than an experimental target-identification study.

CuC7 produced its most negative docking score with PTGS2, whereas AKT1 produced the most negative comparative MM/PBSA estimate. The distinction is important because docking scores and MM/PBSA values arise from different approximations and should not be presented as interchangeable measures of affinity. Redocking of the seven co-crystallized reference ligands yielded RMSDs between 0.186 and 1.490 Å, showing that the selected docking setup reproduced each crystallographic pose within the prespecified 2 Å criterion. The reference inhibitors also provided target-specific context: CuC7 did not uniformly outperform the co-crystallized ligands, and the simulations cannot establish equivalence to clinically or experimentally characterized inhibitor activity.

AKT1 emerged as a prominent candidate across the computational analyses. The AKT1–CuC7 trajectory showed comparatively low backbone deviation, sustained predicted hydrogen bonding, and the most negative CuC7 MM/PBSA estimate among the four simulated targets. AKT signaling is central to prostate cancer cell survival and therapeutic resistance and interacts with upstream EGFR, ERBB2, and PIK3CA signaling (Shorning et al., 2020; Miller et al., 2019; Miyamoto and Lee, 2020). Other cucurbitacins have also been reported to alter EGFR/HER2- and PI3K/AKT-related signaling in cancer models (Wu et al., 2019; Klungsaeng et al., 2020; Liu et al., 2019). These precedents make AKT-related signaling a reasonable follow-up direction, but they do not demonstrate that CuC7 binds AKT1 directly.

The DU145 experiments support this cautious interpretation. CuC7 reduced cell viability and AKT1 protein abundance, but a reduction in protein level can result from transcriptional regulation, altered translation, degradation, pathway feedback, or secondary effects of cell stress and death. It is therefore inappropriate to describe the ELISA result as direct AKT1 inhibition or target validation. Direct binding or target engagement would require experiments such as cellular thermal shift assays, drug-affinity-responsive target stability assays, biophysical binding measurements, or suitable chemical-proteomic approaches. A biochemical AKT1 kinase assay would be needed to test functional enzyme inhibition.

The network results also prioritized EGFR, ERBB2, PIK3CA, MDM2, BCL2L1, and PTGS2. EGFR, ERBB2, and PIK3CA are connected to AKT and MAPK signaling in prostate cancer (Adediran et al., 2022; Miller et al., 2019). MDM2 and BCL2L1 participate in p53 and mitochondrial survival networks, respectively, while PTGS2 links inflammation to tumor progression (Fujita et al., 2002; Vogler et al., 2025). The computational results are therefore compatible with a broader survival, growth-factor, apoptosis, and inflammatory hypothesis. They do not show that these proteins are simultaneously engaged in DU145 cells. Target-specific biochemical and cellular studies are needed before using the term polypharmacology in a demonstrated rather than proposed sense.

The comparative MM/PBSA results should also be interpreted within their methodological limits. The calculation is an end-state estimate derived from finite MD trajectories and depends on force-field parameters, conformational sampling, dielectric treatment, and the selected analysis window (Valdés-Tresanco et al., 2021; Verma et al., 2016). Entropy was not included, sampled frames from a trajectory are temporally correlated, and standard deviations across frames do not replace uncertainty estimates from independent simulations. Thus, the reported values contextualize the modeled complexes but should not be equated with experimental dissociation constants or definitive confirmation of binding.

Finally, the use of a cucumber-leaf derivative links this work to the broader prospect of valorizing agricultural by-products (Bauleth et al., 2025). That sustainability argument remains secondary to establishing reproducible compound provenance, pharmacology, selectivity, and safety. Future work should therefore combine rigorous chemical characterization with orthogonal target-engagement

assays, broader prostate cancer models, normal-cell controls, and in vivo exposure and tolerability studies.

### 4.1 Study limitations

Several limitations constrain interpretation. First, target prediction and network centrality depend on database coverage, curation, literature density, threshold selection, and network topology; highly studied proteins can be preferentially recovered. Second, docking produces model-dependent poses and scores rather than measured affinity. Although crystallographic redocking yielded RMSDs below 2 Å for all seven binding sites, this validation demonstrates recovery of the crystallographic poses under the selected protocol and does not establish experimental binding affinity or target engagement. Third, PTGS2 simulations used the murine 3LN1 structure, introducing a species and construct limitation for a human prostate cancer hypothesis. Fourth, the MD simulations sample finite trajectories and depend on force-field, ligand-parameter, protonation, equilibration, and replicate choices. Fifth, MM/PBSA is an approximate trajectory-dependent estimator; entropy was omitted and frame-wise standard deviations do not fully describe sampling uncertainty. Sixth, biological assessment used one androgen-independent cell line without a non-malignant prostate control, and the AKT1 ELISA comprised a single experiment with three technical replicate wells per group rather than independent biological replicates. The ELISA findings are therefore descriptive and require independent replication. No direct target-engagement assay, kinase assay, or measurement of the other predicted targets was performed. Finally, no pharmacokinetic, toxicity, or in vivo validation was performed. The proposed multi-target interpretation should therefore be treated as a hypothesis requiring independent experimental testing.

## 5 Conclusion

CuC7 reduced DU145 cell viability and was associated with lower AKT1 protein abundance. Network analysis, docking, MD simulations, and MM/PBSA estimates prioritized AKT1, MDM2, PTGS2, and EGFR as candidates for further investigation, while reference inhibitors and crystallographic redocking provided additional context for the computational workflow. Together, the results support a plausible multi-target hypothesis and identify AKT1-related signaling as a priority for follow-up. They do not establish direct AKT1 engagement, kinase inhibition, or simultaneous binding to the proposed target set. Biochemical binding and activity assays, cellular target-engagement studies, additional prostate cancer and non-malignant models, and in vivo studies are required to test the proposed mechanism and evaluate the therapeutic potential of CuC7.

# Abbreviations

| | |
|---|---|
| **AKT1** | RAC-alpha serine/threonine-protein kinase |
| **BCL2L1** | B-cell lymphoma 2-like 1 |
| **CCK-8** | Cell Counting Kit-8 |
| **CTD** | Comparative Toxicogenomics Database |
| **CuC** | Cucurbitacin C |
| **CuC7** | Cucurbitacin C7 |
| **EGFR** | Epidermal growth factor receptor |
| **ELISA** | Enzyme-linked immunosorbent assay |
| **GO** | Gene Ontology |
| **KEGG** | Kyoto Encyclopedia of Genes and Genomes |
| **MAPK** | Mitogen-activated protein kinase |
| **MD** | Molecular dynamics |
| **MDM2** | Mouse double minute 2 homolog |
| **MM/PBSA** | Molecular mechanics/Poisson–Boltzmann surface area |
| **OMIM** | Online Mendelian Inheritance in Man |
| **PCa** | Prostate cancer |
| **PDB** | Protein Data Bank |
| **PI3K** | Phosphoinositide 3-kinase |
| **PTGS2** | Prostaglandin-endoperoxide synthase 2 |
| $R_g$ | Radius of gyration |
| **RMSD** | Root mean square deviation |
| **RMSF** | Root mean square fluctuation |

# Declarations

**Ethics approval and consent to participate.** Not applicable.

**Consent for publication.** Not applicable.

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**Competing interests.** [**AUTHOR QUERY:** Confirm the competing-interest declaration for both authors.]

**Author contributions.** [**AUTHOR QUERY:** Provide a CRediT-style contribution statement for Anthony Peter and Sarah Profess.]

**Data availability.** [**AUTHOR QUERY:** State where the target lists, network files, docking configurations and poses, MD parameter files and starting structures, analysis scripts, MM/PBSA input/output, and numerical source data for all figures can be accessed.]

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